Fleet-level paired gains with 95% t intervals; > 0 favours PI-DSSDM (LSTM gains in Table_paired_tests.csv)

planned duty

ALL | Basis hybrid
ALL | Bayesian PF
ALL | Inverse PINN
ALL | Physics SSM
matched | Basis hybrid
matched | Bayesian PF
matched | Inverse PINN
matched | Physics SSM
discrepancy | Basis hybrid
discrepancy | Bayesian PF
discrepancy | Inverse PINN
discrepancy | Physics SSM
emission mismatch | Basis hybrid
emission mismatch | Bayesian PF
emission mismatch | Inverse PINN
emission mismatch | Physics SSM
combined mismatch | Basis hybrid
combined mismatch | Bayesian PF
combined mismatch | Inverse PINN
combined mismatch | Physics SSM
weak gap | Basis hybrid
weak gap | Bayesian PF
weak gap | Inverse PINN
weak gap | Physics SSM
load exponent | Basis hybrid
load exponent | Bayesian PF
load exponent | Inverse PINN
load exponent | Physics SSM

-4 -2 0 2 4 6 8 10

Paired MAE gain (h), comparator - KI-SSDM

probabilistic duty

ALL | Basis hybrid
ALL | Bayesian PF
ALL | Inverse PINN
ALL | Physics SSM
matched | Basis hybrid
matched | Bayesian PF
matched | Inverse PINN
matched | Physics SSM
discrepancy | Basis hybrid
discrepancy | Bayesian PF
discrepancy | Inverse PINN
discrepancy | Physics SSM
emission mismatch | Basis hybrid
emission mismatch | Bayesian PF
emission mismatch | Inverse PINN
emission mismatch | Physics SSM
combined mismatch | Basis hybrid
combined mismatch | Bayesian PF
combined mismatch | Inverse PINN
combined mismatch | Physics SSM
weak gap | Basis hybrid
weak gap | Bayesian PF
weak gap | Inverse PINN
weak gap | Physics SSM
load exponent | Basis hybrid
load exponent | Bayesian PF
load exponent | Inverse PINN
load exponent | Physics SSM

-4 -2 0 2 4 6 8

Paired MAE gain (h), comparator - KI-SSDM